

EARTHCRAFT VIRGINIA

EarthCraft Virginia serves as a blueprint for building multifamily homes with high energy efficiency and healthy indoor air quality through environmentally responsible design and construction.

BY DEBORAH RIDER ALLEN



Carter Woods II in Richmond was the first affordable housing multifamily project to get EarthCraft certification in Virginia in November 2006. The project was developed and is managed by the Better Housing Coalition.

Healthy, environmentally conscious, energy-efficient homes for low- to moderate-income families? It sounds like an oxymoron. But the EarthCraft Virginia multifamily program is making this concept a reality. EarthCraft Virginia serves as a blueprint for building multifamily homes with high energy efficiency and healthy indoor air quality through environmentally responsible design and construction. It is an adaptation of the EarthCraft green builder program developed ten years ago in Atlanta, Georgia, by Southface Energy Institute and the Greater Atlanta Home Builders Association. (For more on the EarthCraft program,

see “Creating Green Communities,” *HE* Nov/Dec ’05, p. 24.)

For each project, EarthCraft Virginia performs an energy model, meets with all of the principals involved in the administration and construction of the project, and provides the principals with a menu of items they can choose from to get the 200 points required to earn certification for their project. These items fall into various categories, including site planning and landscaping, envelope system, air sealing, insulation, windows, recycled and natural materials, lighting, heating and cooling, indoor air quality (IAQ), ventilation, moisture control, recycling of construction waste, and durability.

An EarthCraft technical advisor is assigned to each project to advise the building team on the building project and to perform quality assurance inspections during construction to verify compliance. The cost of the certification program is quoted at the beginning of the project. Cost varies with the scope and size of the project; it averages \$300–\$500 per unit.

EarthCraft Thrives in Virginia

To bring this program to Virginia, EarthCraft Virginia founders Karl E. Bren and Charles B. Bowles helped facilitate a partnership between the Virginia Community Development Corporation (VCDC) in Richmond

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and Southface Energy Institute in Atlanta, founder of the EarthCraft program, using the EarthCraft Atlanta multifamily program as a template. (VCDC is a nonprofit that provides capital for low-income tax credit deals.) VCDC agreed to be the host organization for the program. "It was a dream come true for us because it meant having a green builder program that would have the blessing of a state agency that instantly would create customers," says Bren, who along with Bowles has been working for the past eight years in green and energy-efficient building with VCDC projects, training, and conferences.

"But it wasn't until Karl envisioned this multifamily effort and we brought the Southface program to Virginia that it really took off," adds Bowles. Katherine C. McGurran, of Southface Energy Institute, came to Virginia to serve as operations director for EarthCraft Virginia. Bowles is program technical director and Bren serves as program director.

A real springboard to the immediate success of the EarthCraft Virginia

program was the Virginia Housing and Development Authority (VHDA)'s adoption of EarthCraft as a part of its Qualified Application Plan (QAP) for low-income project tax credits for developers and builders. Developers applying for tax credits who agree to get certified under the EarthCraft program get 15 points on their score—a critical amount for a very competitive process. Total scoring varies by location, but 900 would be a perfect score. A typical score is usually between 400 and 550 points.

James M. Chandler, director of the low-income housing tax credit program for VHDA, says that his organization decided to adopt the EarthCraft Virginia concept for several reasons. VHDA felt that green building construction "just makes good sense" environmentally, and that the EarthCraft program provides a better living environment and lower utility costs for tenants. Bren says that to his knowledge, this is the first time a state-administered agency has put a specific green builder program into its QAP.

Chandler says that it was a natural jump to add the EarthCraft certification to its QAP, which already included a variety of energy-saving categories. Some of those categories are higher SEER ratings for heat pumps (10 points); Energy Star windows (5 points); exterior walls with brick, a durable, earth-friendly product that is sustainable for the life of the building (20 points); Energy Star appliances (5 points); low-flow faucets and showerheads (3 points); and a water meter within each unit (5 points). Finally, 10 points are awarded to the developer for having a member of the architectural design team who is LEED accredited.

In 2006, 13 out of 35 developments that received tax credits selected the EarthCraft certification. In 2007, 13 out of the 35 that are receiving tax credits selected EarthCraft certification—more than 40%. Chandler adds that while EarthCraft certification will still be worth 15 points in 2007, VHDA is considering increasing the credit to 25 or 30 points in 2008 to encourage builders

EarthCraft in Standard Multifamily Buildings

Not only is EarthCraft Virginia penetrating the multifamily affordable housing market, but it is also permeating standard multifamily residential developments in the state. Scandia USA, a development and building company, is currently building 41 town homes as a part of a multiuse development called Rocket's Landing, in Richmond, Virginia.

"I originally thought building under EarthCraft certification would separate us from the crowd, but after reading several books on the environment, I believe if we build with less impact on the environment then it is better for everyone," says

Roger Petersen, owner and president of Scandia USA.

When looking into green building for the Rocket's Landing project, Petersen called Southface Energy Institute in Atlanta, Georgia, to find a third-party verifier. Southface referred him to the EarthCraft Virginia program.

"I looked at the program and it is above the standards of Energy Star, but a little below the standards of the U.S. Green Building Council, which has very hard-to-follow guidelines. EarthCraft is also more affordable," says Petersen, adding that one of the first options Scandia chose to utilize for points was building on existing land that was previously industrial space. This option allowed the use of existing infrastructure and did not require

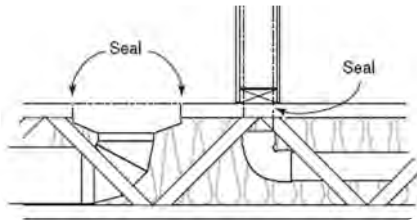
that any vegetation be cleared before construction.

Some of the other options that Scandia chose for the Rocket's Landing project were sealed ductwork for HVAC systems (all of which will be Duct Blaster tested to get the leakage down as close as possible to 5% of total fan flow); blown-in cellulose insulation; high-energy-efficiency-rated Bonneville windows from Canada; and gardens on all the rooftop units. Petersen says he predicts the extra cost to be less than 1% of the hard cost of building.

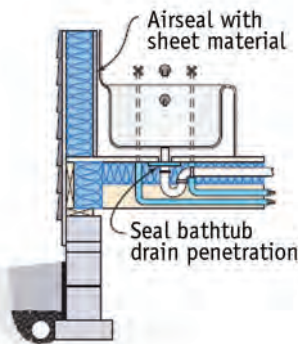
"We did not do it for tax credits, but I just found out a few weeks ago that I can apply for tax credits, so that was a bonus," says Petersen. "Really, I just think it is the right thing to do."

Carter Woods II Construction Details

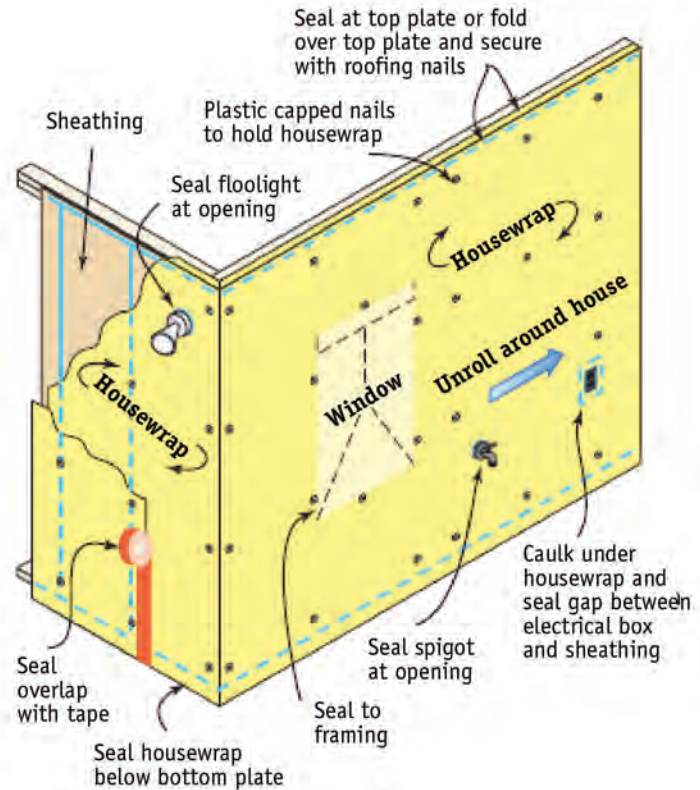
The following graphics show some of the things that were done on the Carter Woods II project to get EarthCraft Virginia certification. These graphics are all included in the EarthCraft House building manual.



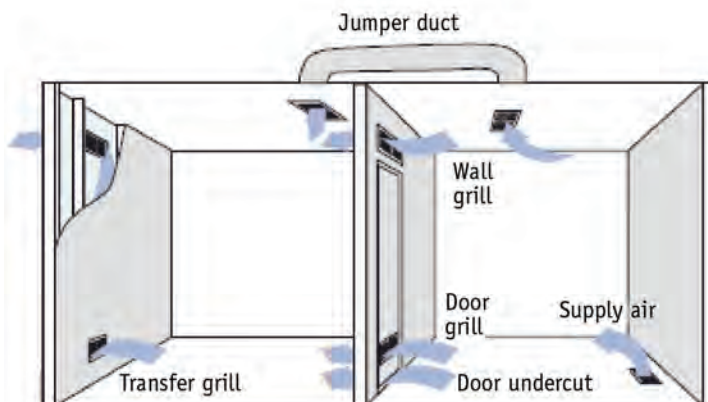
1. Penetration through insulated subfloor sealed. All penetrations through insulated floor systems over unconditioned areas (such as basements, crawlspaces, and garages) must be sealed, including duct, electrical, and plumbing penetrations.



2. Shower and tub drain sealed. Plumbing penetrations shall be blocked with sheathing and sealed at edges with proper sealant. Rock wool or similar product does not qualify.



3. House wrap installed with all seams and edges taped. House wrap shall be installed according to the manufacturer's specifications. All seams must be cut properly and all edges taped to ensure a continuous air barrier.



4. Multiple return ducts or transfer grilles. Each bedroom shall have a dedicated return duct or, for homes with no return ducts located in bedrooms, all supply air shall have a direct path back to a return grille even when doors are closed. This path shall be through transfer grilles. Houses with undercut doors to bedrooms do not qualify for these points.

and developers to pursue EarthCraft certification.

Bren adds that the projects that receive credits and commit to EarthCraft are managed by professionals, who are highly motivated to follow through, since VHDA requires that all QAP commitments be met. Failure to do so brings penalties in future applications. Virginia has a very competitive process when it comes to applying for tax credits, and a few extra points could mean the difference between receiving a credit and being turned down.

The EarthCraft program is changing more than just affordable multifamily housing in Virginia. Market-rate developers and builders in the state are also looking for the benefits of the EarthCraft designation (see “EarthCraft in Standard Multifamily Buildings,” p. 19).

Teaching by Example

To showcase the process and prove that it works, EarthCraft Virginia spearheaded three affordable housing pilot projects during 2006. Cassell Pines, in Wytheville, is 18 new apartment units. Yorktown Square II, in Yorktown, is the renovation of 60 apartment units. And Carter Woods II, in Richmond, is 71 new apartment units; certified in November 2006, it was the first affordable housing multifamily project to get EarthCraft certification in Virginia.

The Better Housing Coalition (BHC), a nonprofit developer that focuses on building affordable mixed-income green sustainable communities is the developer and manager of Carter Woods. Bob Newman, director of operations for BHC, says that the jump to the EarthCraft Virginia certification was not that big for his organization. That was because BHC had already

built the first phase of Carter Woods according to some green building principles when it decided to do the second phase as part of the EarthCraft pilot program.

“As a corporation we have been moving more and more towards environmentally friendly elements in design and construction, so when we built phase I of Carter Woods it



Carter Woods II consists of 71 affordable housing units. Each unit has Energy Star dishwashers and refrigerators.

had a lot of elements in it that were higher-quality materials. So when we looked at doing phase II and seeking this EarthCraft certification, we did not have to make that many changes,” says Newman.

One of the biggest changes BHC did make in Carter Woods II was to the mechanical design. Through energy modeling, EarthCraft identified a host of air sealing techniques that tightened up the building so much that it had to be mechanically ventilated. That in turn caused a slight change in the architectural design to accommodate the mechanical ventilation system. BHC also had Duct Blaster tests conducted to verify the tightness of the duct systems. This improved the efficiency of the systems and lowered the residents’ utility bills. (For more on the Carter Woods II project, see “Carter Woods II Construction Details.”)

Other selections that BHC made for its EarthCraft certification on Carter Woods II included changing the HVAC refrigerant to environmentally friendly Puron (R-410A), and using a low-volatile-organic-compound (VOC) carpet. The backing on this carpet is made from the recycled material that keeps automobile glass from shattering. It used blown-

cellulose wall insulation, Energy Star dishwashers and refrigerators, low-VOC paints, low-VOC sealant on particleboard, linoleum instead of vinyl for durability, low-flow water fixtures, and high-efficiency outdoor lighting with automatic motion sensor controls. And in every apartment unit, each time the bathroom light is turned on an exhaust fan automatically runs for 15 minutes to exhaust any moisture from the room.

BHC chose to landscape mainly with native trees and shrubs. This allowed it to forgo an irrigation system, lawn maintenance, and the use of harmful fertilizers and pesticides. The resulting environmentally friendly landscape has been certified through the National Wildlife Federation as a wildlife habitat.

An extensive sealing package included installing the house wrap with all seams taped to ensure a continuous air barrier. All penetrations through insulated wall systems were also sealed, as were all plumbing penetrations at showers and drains. And in each bedroom, BHC’s subcontractors installed transfer grilles with a direct path to the return air. Several dumpsters were provided during construction to separate cardboard, wood, and concrete waste on the site; this waste was later recycled.

Stewart Jester, senior project manager for BHC on the Carter Woods job, says that the coalition

had also planned on recycling excess drywall from the job and had a group lined up who would take it after it was ground up to use as a lawn product. But when the subcontractor realized that he would be responsible for moving and grinding up any scrap, he became so efficient with the sheets of drywall that there was very little to be ground up.

be 5% over a standard multifamily package, but it is still a very good investment,” he says.

So far EarthCraft Virginia has already surpassed its Atlanta counterpart in completed, ongoing, and upcoming multifamily projects. “Having an incentive and a competitive process like VHDA does with the tax program, and not forcing anyone to do it, spreads this

quicker than just an organic process,” says Bren. “If you took VHDA out of it you might have some people who’d do it, but not many. We were fortunate that Atlanta and Southface had this program and we adopted it here. Then having an agency like VHDA to

give an incentive to certify it is an advantage.”

“The success of this program demonstrates the ability of other states to take a like approach to greening their residential housing,” says Bowles. “This is not rocket science. Most of it is basic building science that we have known for ten years. It is not the technologies that we are using in the program that are so new; it is that we are developing the program and making an impact on housing.”

“And having a recognized third-party green builder program is an important way to green America,” says Bren.

Deborah Rider Allen grew up in Richmond, Virginia. She writes for businesses and publications including Home Energy, the Richmond Times-Dispatch, Housetrends Magazine, and R-Home Magazine.



The backing on this carpet is made from the recycled material that keeps automobile glass from shattering.

Jester also says that BHC specifically chose items totaling more than the required 200 points so that if something was missed it would not jeopardize the certification. One item BHC did miss was the protection fencing around existing trees. Jester says the subcontractor put orange protection fences around the trunks of the trees when they were supposed to be around the canopy drip line of the tree. “So Earthcraft said it was a good try, but it did not qualify under the rules, so we did not get points for that. But it was a good experience for us because we were just learning and now we know,” says Jester.

Carter Woods II scored 248 points. Newman reports that the cost of these items added about 3% to the cost of the contract. “That being said, phase I was probably a higher quality than most people would be building multifamily, so it might

